Assignment 6; Noah Wiley CSCI 113

1) Two values are compared using slt instruction in 32-bit ALU by subtracting them and examining sign bit of outcome. In event of a negativ outcome, output is set to 1, signifying that initial value is lower. ALU is instructed by control signal 0111 to carry out subtraction and determine if result is negative to produce intended output

2)

C	AC	MD	MQ	
0	0000	0101	1110	Add
0	+1110	↓		Shift } 1st
	1110			
	0111	0101		Shift } 2nd
0	0011	0010		Add } 3rd
	× 1110	1001		Shift
	0001			Shift } 4th
1	0001	1001		
0	1000	1100		
0	0100	0110		

Final Answer: 0100 0110

3) a. 1010011010

1= -1 01 = 1 10 = -1 00 = 0

11 = 0 11 = 0 01 = +1

-1, 1, -1, 0, 1, 0, 0, +1, -1

b.

AC	MQ	MQ-1	MD $\frac{0101}{\times 1}$ 0110
0000	0101	0	1010
+0110			-MD = 0110
0110	0101	0	Add
0011	0010	1	Shift
+1010			
1101	0010	1	Add
1110	1001	0	Shift
+ 0100			
00001	1001	1	
1000	1100	0	
+ 1010	1100	1	
0010	1100	1	Add
10010110			Shift

Final Answer: 1001 0110

4)

MD	AC	MQ	
0100	0000	1011	
	0001	0110	1st
	1101	0110	
	0001	0110	
	0010	1100	
	1110	1100	2nd
	0010	1100	
	0101	1000	
	0001	1001	3rd
	0001	1001	
	0011	0010	
	1111	0010	4th
	0011	0010	
	R=3	Q=2	

5)

MD 0100	AC	MQ	
	0000	1011	
	0001	0110	
	1101	0110	1st
	0001	0110	
	0010	1100	
	1110	1100	2nd
	0010	1100	
	0101	1001	
	0001	1001	
	0011	1001	3rd
	1111	0010	
	0011	0010	
		0010	

2's MD = 1100

0011 (R=3) 0010 (Q=2)

6) The non-restoring division algorithm is more efficient than the restoring division algorithm because it avoids the need to "restore" the partial remainder after each subtraction, resulting in fewer operations per iteration and faster execution, as it only needs to perform a single addition or subtraction based on the sign of partial remainder, rather than potentially needing to add back the divisor in the restoring method.